

Which Arab? How LLMs Rate Bank Customers from the 22 Arab League Countries in Money-Laundering Checks

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Notes: title (not report text; rewrite in your own words)

The title above is a working title proposed by Claude Code on 2026-10-08 at Laith's request. The team has to confirm or replace it. Other options, none of which assumes a result:

- Same File, Different Passport: Auditing LLMs in Bank Customer Checks
- One Label, 22 Nationalities: Do LLMs Tell Arab Bank Customers Apart?
- Which Passport Is High Risk? A Nationality-Swap Audit of LLMs in Anti-Money-Laundering Checks

- Method: 20 synthetic customer files; 28 nationalities, changed in the nationality and passport lines only; a one-page bank policy in two versions; risk score 0–100, rating, enhanced checks, recommendation; four open-weight models.
- Results: nothing has been run as of 2026-10-08. Write this part last, and only from the blocks that were run.
- Last sentence: what follows. The data decide: the models separate the nationalities, lump them together, or follow the list.

Abstract

Notes: abstract (not report text; rewrite in your own words)

At most 250 words. The rubric asks for the research questions, the methods and the results. In all notes, (F) after a source means the full text or the cited passages were read and (A) means only the abstract was read: check the full text before citing a number from it. (press) is a press report and (vendor) a company's own statement.

- Context: banks must check every new customer for money-laundering risk. AI systems are being sold for this check, and EU law regulates their use in it.
- Why Arab customers: LLMs repeat stereotypes about Arabs and Muslims in what they write. Banks in Europe answered enquiries from Arabic-sounding names far less often. Numbers and sources: introduction notes.
- Why this task: the nationality cannot be left out of the file (the bank must record it), and the rules say how far it may count (they speak of residence, not of the passport).
- Gap: every study treats "Arab" as one group. We found no study in which an LLM rates bank customers while their nationality is varied; this rests on a few web searches only.
- Questions (from the proposal of 2026-10-08; hypotheses are not written yet): (1) with the customer file held fixed, do a model's ratings differ across the 22 Arab League nationalities, and by more than across unrelated nationalities? (2) Do the differences follow the EU list of high-risk countries? (3) Arab League nationalities against German, Turkish and placebo nationalities. (4) Does a paragraph saying that nationality is not a risk factor change this? (5) Do the four models agree?

1 Introduction

Notes: motivation, step by step (not report text; rewrite in your own words)

1. **LLMs in decisions about people.** Audits mostly vary race and gender, often through names. Nationality is rare: it appears in 13.2% of 189 papers on LLM bias, gender in 79.9%. In finance: one line stating the applicant's race on a mortgage file; GPT-4 Turbo recommended approval 8.5 percentage points less often and an interest rate 35 basis points higher for Black applicants. Sources: ghosh2025bias (A), bowen2025measuring (F, sections 1–4).
2. **LLMs and Arabs.** In generated text: Muslims tied to violence in 66 of 100 GPT-3 completions; MENA names 78% under-represented in generated workplace stories; Middle Eastern personas described through religion. Models also tell Arab countries apart: Libyan, Sudanese and Tunisian among the five most negative demonyms in GPT-2 text; additional scrutiny suggested for applicants from Iraq, Sudan and Somalia in simulated immigration decisions, while most country effects in the choices were not significant; the values of some Arab countries represented much better than others. Sources: abid2021persistent (F), shieh2026intersectional (F), cheng2023marked (F), venkit2023nationality (F), mao2025gatekeepers (F), zahraei2025menavalues (F).
3. **Banks.** Field experiment, 1,218 e-mails to banks in seven European countries, asking only for contact details: 55.2% replies to domestic-sounding names, 31.6% to Arabic-sounding names; loan enquiries 59.8% vs 35.1%. The Netherlands Institute for Human Rights ruled in 2024 that ING discriminated against two customers whose payments were blocked and checked because of names that did not sound Dutch. Campaigners say HSBC closed dozens of accounts of Syrian nationals living in the UK from 2014. Sources:

stefan2018ethnical (F; the list of countries is in an appendix that was not read, so do not write that Germany is among them), nltimes2024ing (press), mee2016hsbc (press).

- **4. One category.** All of these treat “Arab” or “Arabic-sounding” as one group. A name cannot do better: raters assigned 34% of Moroccan names to Morocco. Source: martiniello2022signaling (F).
- **5. Why a bank’s customer check.** A German bank must record name, place and date of birth, nationality and address of every customer, and rate the money-laundering risk. So the nationality is always in the file. Sources: gwg2017 (F, sections 11 and 15, annex 2).
- **6. The yardstick.** European guidelines: country risk is about where the customer is based or resident, the main places of business, and relevant links. The word “nationality” does not occur in the 225 pages of the final report. An isolated risk factor does not necessarily move a customer into a higher category. The Commission’s list of high-risk third countries has 26 entries; four are Arab League members: Algeria, Lebanon (since 5 August 2025), Syria, Yemen (since 23 September 2016). German law applies the list to relationships and transactions involving such a country or a person resident there. No disadvantage on account of nationality when opening a payment account. No blanket refusal of categories of customers. Sources: eba2021riskfactors (F, guidelines 2.9 and 3.3, word search), ec2026highrisk (F, read 8 October 2026), gwg2017 (F), zkg2016 (F, section 3), eba2023derisking (F, paragraphs 9–10).
- **7. AI in this task.** EU regulation 2024/1624, Article 76(5), applies from 10 July 2027: decisions “resulting from automated processes [...] or from processes involving AI systems” are allowed if accepting or refusing a customer is “subject to meaningful human intervention” and the customer can get an explanation. Vendors sell AI agents for customer checks and screening alerts; one says 10 of the top 20 banks use its agents. ECB: strong increase in AI use cases in banks between 2023 and 2024; for credit scoring and fraud detection mainly decision trees. Sources: eu2024amlr (F, Articles 76(5) and 90), workfusion2026agents (vendor), sardine2026agentic (vendor), ecb2025ai (F).
- **8. Gap and what we do.** No study was found in which an LLM rates bank customers while their nationality is varied. This rests on a few web searches of 8 October 2026, not on a systematic search: write “to our knowledge” only after one has been done. Then the question and one sentence on the design.

Notes: research questions (as in the proposal of 2026-10-08) (not report text; rewrite in your own words)

- Framing: the 22 countries differ in wealth, stability and how their citizens are seen abroad. Does a model carry that into a decision about an individual customer?
- RQ1 (main): do risk ratings and recommendations differ across the 22 Arab League nationalities with the file otherwise identical, and more than across the placebo nationalities?
- RQ2: do the differences follow the EU list of high-risk countries (four of the 22), or a country’s economic and political position?

- RQ3: are Arab League nationalities as a group rated differently from German, Turkish and placebo nationalities?
- RQ4: do the differences remain when the policy says that nationality is not a risk factor?
- RQ5: do the four models agree?

Notes: why it matters (rubric: science, technology, society) (not report text; rewrite in your own words)

- Science: bias audits report one number for “Arab”. Here there is an official yardstick for the 22 countries, so three outcomes can be told apart: the model follows the list, lumps all 22 together, or draws its own lines.
- Technology: AI agents are sold for customer checks (step 7), and EU law expects AI systems in this decision. How language models react to the nationality in a customer file is untested, as far as we found.
- Society: an account is needed for wages, rent and daily life. A higher risk rating means more checks, delays or a refusal. This last sentence is our own argument and has no source.

Notes: do not overstate (not report text; rewrite in your own words)

- Not “banks use LLMs to decide on customers”: what the products do, which models they use and which fields the model sees is not public. We test open models in the task the products are sold for.
- Not “LLMs discriminate against Arab customers” as a fact. Nothing has been run. Explicit origin labels can also make models more cautious (bunel2026customer, F).
- A higher rating for a listed country is not automatically wrong in practice. Our argument is narrower: for a customer who lives and works in Germany and has no payments abroad, the written rules give the passport no weight.
- The legal texts were read in single sections only (see the literature matrix). Nobody on the team is a lawyer. Do not present the policy as legal advice.
- Say “Arab League nationalities”, not “MENA” and not “Arab customers”.
- The claim to be first needs a systematic search that has not been done.
- Do not cite numbers from mohammad2026mirage: they are placeholders.
- This is an audit of model behaviour, not a tool for checking customers. The risk score is not a probability.

2 Related Work

Notes: what each strand teaches us (not report text; rewrite in your own words)

The rubric asks for what is relevant to our work, not for a list of papers.

- **LLM audits with one explicit line.** Mortgage files with a race line: gaps in approval and interest rate, larger for weak files; the rate gap in all eight models. Llama 3 8B approved every application, so approval could not be analysed for it. Claude 3 Opus answered only 74% of the requests for Black applicants (bowen2025measuring, F). For us: a yes/no or three-level answer may not move in small models, hence the 0–100 score; count refusals by nationality.
- **Direction is unstable.** Real origin labels led to fewer discriminatory recommendations than invented groups (bunel2026customer, F); every detected race or ethnicity bias favoured the minority in one study (arcuschin2026blind, F). For us: expect either direction.
- **Explicit cues.** A stated attribute moves decisions more than a name (tamkin2023evaluating, F). Different cues for the same group can give different conclusions (tonneau2026cues, A). For us: the results hold for a nationality and passport line, not for names.
- **Nationality in LLM decisions.** 20 nationalities in job recommendations, Jordan the only Arab one (salinas2023unequal, F). GPT-4 prefers developers from some countries, tested by swapping only the location string (nakano2024nigerian, F). Country effects mostly not significant in simulated immigration choices (mao2025gatekeepers, F).
- **Arab group in LLM studies.** Always pooled: lippens2024computer (F), hoffmann2026evaluating (F), bai2025explicitly (A), albaroudi2026addressing (F). These are hiring studies; none is about banking.
- **Telling the model to ignore the attribute.** Brought discrimination close to zero in Claude 2.0 (tamkin2023evaluating, F). A “use no bias” sentence removed the approval gap in mortgage files (bowen2025measuring, F). Instructions fail once realistic context is added in hiring (karvonen2025robustly, F). For us: the second policy version can go either way.
- **Models may notice a test.** vohra2026audit (F), needham2025large (A).
- **Human evidence in banking.** See the introduction notes, step 3.
- **Not searched yet.** Research on bias in anti-money-laundering systems and on “de-risking”. Find sources, or say that this literature was not covered.

3 Data

Notes: materials (not report text; rewrite in your own words)

- 20 customer files for a current account, one per occupation (the 20 occupations of the earlier hiring design). Net incomes are invented.

- Held constant in every file: lives in Germany, tax resident in Germany, employed by a German employer, salary as source of funds, no payments to or from other countries, no sanctions match, not politically exposed, no adverse media.
- 10 plain files, 10 with one mild indicator named in the policy: account opened by video call (4), cash deposits from tips (2), second income in cash (2), second income without cash (2).
- **Nationality:** 28 conditions. 22 Arab League nationalities, 2 benchmarks (German, Turkish), 4 placebo nationalities (Uruguayan, Malawian, Maldivian, Cambodian). No version without a nationality: the field is mandatory.
- **What changes with the nationality:** the nationality line and the passport line. All non-German customers hold an unlimited settlement permit. The German customer has an identity card and is a citizen, so that file differs in three lines.
- **The policy:** one page, a paraphrase of public rules, with the Commission’s list of 26 high-risk countries as of 29 January 2026. Second version: one more paragraph saying that nationality is not a risk factor.
- **Country data:** EU list status in customer_nationalities.csv; World Bank income and region in country_covariates.csv.
- Everything is synthetic: no real customer, bank or employer. The bank’s name (Lindenufer Bank) is invented.
- Still needed for the rubric: descriptive statistics and plots of the answers.

Source: the README files under experiment/.

Notes: what a customer file looks like (not report text; rewrite in your own words)

The file shows a reference number instead of a name. Example of the lines that change:

```
Nationality: Jordanian
Identity document: Jordanian passport,
checked in the original
Residence status: Settlement permit
(Niederlassungserlaubnis), unlimited
```

- A name cannot tell Arab countries apart, and it adds signals we do not want (gender, religion).
- A real file also holds the place of birth. It is left out so that the nationality is the only link to another country. Say so as a limitation.

Notes: nationalities whose Arab identity is contested (not report text; rewrite in your own words)

Comoros, Djibouti and Somalia are Arab League members whose Arab identity is contested. They stay in every comparison. Point them out as especially interesting cases and look at where they fall in the results.

4 Methods

Notes: experiment setup (not report text; rewrite in your own words)

- One run: a main prompt (the model assists the onboarding team of a bank in Germany; an analyst takes the final decision; the policy), then a message with one customer file and the request.
- Answer: a JSON object with a risk score 0–100, a rating (low, medium, high), enhanced checks yes/no, a recommendation (open, escalate, decline) and a short reason.
- Two policy versions: without and with the paragraph on nationality.
- Three wordings of the request.
- Five repetitions per prompt are planned; the decoding settings are not decided.
- Models: Qwen3 8B, Llama 3.1 8B, Gemma 3 12B, Mistral Small 3.2 24B. State the exact versions and settings.
- `scripts/build_runs.py` builds and checks the prompts. It calls no model. The code that sends the runs does not exist yet.
- How the answers are analysed: not decided as of 2026-10-08.

Planned runs per model:

Block	Runs
1 Policy without the nationality paragraph	8,400
2 Policy with the nationality paragraph	8,400
Total	16,800

Each block: 20 files $\times$ 28 nationalities $\times$ 3 wordings $\times$ 5 repetitions. Source: `experiment/README.md`.

5 Results

Notes: nothing to report yet (not report text; rewrite in your own words)

As of 2026-10-08 no run exists. The proposal lists what to compare:

- the spread of risk scores and ratings across the 22 Arab League nationalities, against the spread across the 4 placebo nationalities;
- the four listed countries against the eighteen others;
- unlisted Arab League nationalities against placebo, German and Turkish;
- with versus without the paragraph on nationality;
- plain files against files with one indicator;
- whether the four models agree;
- how often a model refuses or gives an unreadable answer, by nationality.

The hypotheses and the analysis have to be written down before the first real run. A pilot has to show first whether the answers vary at all. Block 1 on all four models is the plan for the midterm.

6 Conclusion and Discussion

Notes: alternative explanations to discuss (not report text; rewrite in your own words)

- A passport also says “foreigner”. The German file is the only one of a citizen, and it differs in three lines. Compare the Arab League nationalities with the Turkish and placebo files, which are foreigners too.
- The policy names the listed countries. A model may apply a rule about residence to a passport because the country name appears in the prompt. A run without the list would separate this; it is not planned.
- The label may stand for religion, conflict or income of the country, not for the nationality as such. `country_covariates.csv` has income and region.
- Any set of labels spreads the scores a little. The placebo nationalities are the yardstick; with only four of them it is a rough one.
- Models may notice that this is a test (`vohra2026audit`, `F`; `needham2025large`, `A`).
- All 20 customers are ordinary. If a model answers “low” for everybody, the rating cannot show a difference; the score may.
- A result of “no difference” is an answer too.

Limitations

Notes: limits to state (not report text; rewrite in your own words)

- **FILL IN AFTER THE RUNS:** which blocks and which models were actually run. As of 2026-10-08 nothing has been run.
- The products in this field run on large commercial models and their prompts are not public. We test open models of 8 to 24 billion parameters with a policy and a file that we wrote.
- In practice most customer risk ratings come from fixed scorecards; language models review cases and write the reasoning. We imitate that step.
- The policy is a paraphrase by non-lawyers of rules read in part.
- English prompts for a German bank; one bank, one product, one country.
- Synthetic files without name and place of birth; nationality given as explicit lines.
- One list (the Commission’s) on one date. Other lists differ and change.

Author Contributions

Use of AI Tools

Notes: what to declare (not report text; rewrite in your own words)

As of 2026-10-08, assisted by Claude Code: literature search and research notes; the search for and comparison of scenarios (research/scenario_reviews/); design documents; the bank policy text, the customer files and the prompts; the first drafts of both project proposals (2026-10-04 and 2026-10-08); repository code and structure; the working title and title suggestions; and these notes. For the earlier hiring design also the job ads, CVs and prompts, which are no longer used. Keep the list up to date.

Declaration

A Prompts

Notes: prompts (not report text; rewrite in your own words)

Copy in, unchanged, the main prompt, both versions of the policy and one wording of the request from experiment/prompts/ and experiment/policy/, with one customer file.